

Asmit Gupta

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SUMMARY

Backend-focused Computer Science student seeking to design and scale high-performance, secure systems in production environments.

EDUCATION

Vellore Institute of Technology, Vellore

2024 - 2029

Integrated M.Tech in Computer Science and Engineering

CGPA: 9.47/10.0

Birla Vidyamandir, Nainital

2023

Class 12th (CBSE)

Percentage: 94.6%

WORK EXPERIENCE

Ealth Technologies

January 2026 - Present

Software Development Engineer Intern

- Optimized search API by eliminating N+1 query issue using JOIN LOAD in SQLAlchemy, reducing database round trips and average response time from ~180ms to ~100ms.
- Optimized signup flow by moving OTP email sending to async post-commit, reducing failure response time by ~99%.
- Implemented secure authentication flows using JWT and OAuth2, enabling stateless login and role-based access control.

Society for Industrial and Applied Mathematics (SIAM), VIT

March 2026 - Present

Technical Head

- Oversee technical operations including code reviews and conducting sessions on GitHub, Web Development, and Competitive Coding
- Developed backend infrastructure for Math Premier League handling 200+ concurrent participants with real-time updates – [GitHub](#)

PROJECTS

QueryShield – Rate Limiting & Caching Middleware

FastAPI, Redis – [GitHub](#)

- Built production-grade middleware with adaptive threshold-based caching and sliding 1-hour window for hot endpoint detection
- Achieved 98.6% cache hit ratio (361 DB hits out of 30,047 requests) with 33× average speedup (10ms vs 372ms response time)
- Blocked 86% attack traffic in single-IP DDoS simulation while maintaining 14% throttle rate for legitimate multi-IP traffic
- Sustained 250 RPS burst load with zero 500 errors, delivering 11ms median latency under stress testing with Locust

Hackmates – Modular Backend System

FastAPI, PostgreSQL – [GitHub](#)

- Built scalable REST APIs with clean layered architecture and cross-cutting middleware for logging and security
- Optimized database queries achieving 2-5ms execution time; delivered 120-180ms avg response under 100+ concurrent users

TECHNICAL SKILLS

Backend

FastAPI, RESTful APIs, SQLAlchemy, WebSocket

Security & Performance

OAuth2, JWT, Redis (Caching), K6, Locust (Load Testing)

Databases

PostgreSQL, MySQL

Languages & Tools

Python, Java, Git, Docker, Postman

ACHIEVEMENTS

- Academic Excellence:** Ranked 6th overall among peers with a CGPA of 9.47